GOOGLE CLOUD · PROFESSIONAL CLOUD ARCHITECT

Thinking like the exam: architecture, case studies and key security

Designing & planning · Agent Platform · Analyzing & optimizing processes · Altostrat Media, Cymbal Retail, EHR Healthcare, KnightMotives · Cloud KMS, CMEK & Secret Manager

THE EXAM

What you are walking into

~25%
of the exam is Section 1: designing & planning

2 of 4
case studies appear on each exam

50–60
multiple-choice & multiple-select questions

2 hrs
to finish; case studies open on split screen

The six sections of the guide

1 **Designing & planning a cloud solution architecture**

2 Managing & provisioning a solution infrastructure

3 **Designing for security & compliance**

4 **Analyzing & optimizing technical and business processes**

5 Managing implementation

6 Ensuring solution & operations excellence

Highlighted = the sections this deck focuses on

EXAM TECHNIQUE

Read every scenario in four passes

1 → 2 → 3 → 4

Business goal

What outcome does the company want? Cost, speed, compliance, new revenue, customer experience.

Hard constraint

Find the non-negotiable: RTO/RPO, data residency, HIPAA, latency, existing skills, budget.

Eliminate

Drop answers that add ops burden, break a constraint, or solve a problem nobody asked about.

Pick the Google way

Prefer managed, least-privilege, least-ops, cost-aware — the option Google would publish as a reference.

Keyword reflexes

"minimal operational overhead"
→ **managed / serverless**

"global, strongly consistent"
→ **Spanner**

"lift and shift quickly"
→ **Compute Engine + Migrate to VMs**

"control and revoke keys"
→ **CMEK with Cloud KMS**

"private, high-bandwidth to on-prem"
→ **Dedicated Interconnect**

"ground answers in our data"
→ **RAG on Agent Platform**

Translate requirements into an architecture

Business requirements

- Business use cases and product strategy
- Cost optimization: CapEx → OpEx, TCO
- Compliance and data residency
- Success measurements (KPIs, ROI)
- Time to market, existing skills

Technical requirements

- High availability and failover design
- Elasticity and scalability to demand
- Latency targets and user location
- Backup, recovery, RTO / RPO
- Integration with on-prem and other clouds

The answer the exam rewards

Meets every stated requirement with the fewest moving parts: managed services, least privilege, a clear migration path, and Well-Architected Framework thinking (ops excellence, security, reliability, cost, performance).

Map the workload to the right compute

Scenario cue	Pick	Why it wins
Lift-and-shift VMs, custom OS or licensing	**Compute Engine + MIGs**	Full control; autoscaling and autohealing via managed instance groups
Containerized microservices, portability, many clusters	**GKE (Autopilot / Standard)**	Orchestration at scale; fleets manage multiple environments consistently
Stateless HTTP service, spiky traffic, scale to zero	**Cloud Run**	Serverless containers, pay per request, least operations
Event-driven glue (file lands, message arrives)	**Cloud Run functions**	Triggered by Pub/Sub, Cloud Storage or Eventarc events
Model training / inference at scale	**GPUs / TPUs on GKE or Agent Platform**	Managed accelerators and MLOps tooling
Factory floor, vehicle depot, disconnected or sovereign site	**Google Distributed Cloud**	Google Cloud services running on-prem or at the edge

Choose storage by access pattern, not habit

Requirement in the question	Service
Unstructured media, backups, data lake; cold content cost	**Cloud Storage (+ Autoclass)**
Regional relational: MySQL, PostgreSQL, SQL Server migration	**Cloud SQL**
High-performance PostgreSQL, mixed transactional + analytics	**AlloyDB for PostgreSQL**
Global relational, strong consistency, horizontal scale	**Spanner**
High-throughput time series, IoT / vehicle telemetry	**Bigtable**
Mobile / web document data, offline sync	**Firestore**
Analytics warehouse, SQL ML, data sharing	**BigQuery**
Cache or session store (Redis)	**Memorystore**

Connect, move and protect

Connectivity

- Dedicated Interconnect: private, high bandwidth, SLA with redundancy
- Partner Interconnect: no colocation presence
- HA VPN: fast to set up, encrypted, 99.99% SLA
- Shared VPC: central network control
- Private Service Connect for private access to services

Migration path

- Assess → plan → migrate → optimize
- Migrate to VMs for servers
- Database Migration Service for MySQL, PostgreSQL, SQL Server
- Storage Transfer Service or Transfer Appliance for bulk data
- Global load balancing + Cloud Armor + Cloud CDN at the edge

DR patterns: RTO / RPO vs cost

Backup & restore
Hours · lowest cost

Cold standby
Hours · low cost

Warm standby
Minutes · moderate

Hot / active-active
Near zero · highest

Gemini Enterprise Agent Platform (formerly Vertex AI)

Naming status

Vertex AI was rebranded as Gemini Enterprise Agent Platform at Google Cloud Next, April 2026. APIs and endpoints did not change.

Exam guides and practice questions may still say “Vertex AI”. Treat the names as the same platform.

Before test day, re-check the official exam guide for the product names it uses.

Build

- Gemini and partner models via Model Garden
- Agent Development Kit (ADK)
- Prompting, tuning, grounding

Scale

- Agent Engine managed runtime
- Endpoints for model serving
- Pipelines and MLOps

Govern

- Agent Registry for inventory
- IAM, VPC Service Controls, CMEK
- Model Armor for prompt/response safety

Optimize

- Evaluation of model and agent quality
- Monitoring and observability
- Cost and latency tuning

AI answers the exam rewards

When the scenario says…	Choose	Watch out for
Answers must use our documents or catalog	**RAG: enterprise search / vector search + Gemini**	Training a custom model first
Summarize, tag or transcribe audio and video	**Gemini multimodal (+ Speech-to-Text)**	Building manual tagging pipelines
Reduce call-center load, 24/7 self-service	**Conversational agent with human handoff**	No escalation path
Protect sensitive data in prompts	**Sensitive Data Protection, Model Armor, VPC-SC, CMEK**	Sending PHI/PII unfiltered
Trust, accuracy, brand safety	**Grounding, evaluation, human-in-the-loop review**	Auto-publishing AI output
Limited ML team	**Foundation model + prompt/RAG → tune → train last**	Starting from scratch

Optimize the software delivery lifecycle



Plan	Build	Test	Deploy	Operate	Learn
Requirements, architecture decision records	Cloud Build, Artifact Registry, IaC with Terraform	Unit, integration, load and security testing	Cloud Deploy: canary, blue/green, rollback	Cloud Monitoring, SLOs, error budgets, alerting	Blameless postmortems, root-cause fixes

Exam signals

- “Releases break production” → canary + automated rollback
- “Alerts by email are ignored” → alerting policies, on-call, SLO burn rate
- “Each team has its own pipeline” → standardized CI/CD

Reliability & troubleshooting

- Chaos and DR testing on a schedule
- Cloud Logging + Trace + Profiler to find root cause
- Measure with SLIs; spend error budget on change

Optimize how the business adopts the cloud

Stakeholder management

Influence and facilitate; align sponsors early

Change management

Phased rollout, training, clear communication

Team assessment

Skills readiness; training or partners to close gaps

Decision-making

Document trade-offs; data-driven choices

Customer success

Measure adoption and outcomes, not just uptime

Cost optimization

CapEx → OpEx; CUDs, Spot VMs, rightsizing, budgets

Business continuity

DR plans, tested runbooks, RTO/RPO agreed with business

Exam tip

Pick answers that involve people and process, not only technology

Four companies, four architecture stories

EH

EHR Healthcare

Healthcare SaaS

Exit colocation, containers to GKE, HIPAA, 99.9% availability, hybrid insurer integrations

CR

Cymbal Retail

Online retail

Gen AI catalog enrichment, better product discovery, conversational self-service

AM

Altostrat Media

Media & content

Already cloud-native; layer gen AI for summaries, moderation, recommendations

KM

KnightMotives

Automotive

Connected and autonomous vehicles, hybrid edge, data sovereignty, data monetization

Migrate a regulated SaaS out of colocation

What they need

- Leave colocation before a lease expires
- 99.9% availability for customer-facing systems
- Stay compliant (HIPAA) while cutting admin cost
- Onboard new insurance providers faster
- Centralized monitoring, logging and alerting
- Insights and predictions on healthcare trends

Requirement	Google Cloud answer
Containerized apps, many environments	**GKE with fleet management**
MySQL, SQL Server, Redis, MongoDB	**Cloud SQL, Memorystore, MongoDB Atlas / Firestore**
Secure, fast link to on-prem	**Dedicated Interconnect (redundant)**
Legacy insurer integrations	**Hybrid connectivity now; Apigee for APIs**
Microsoft Active Directory users	**Managed Microsoft AD or Cloud Identity sync**
Email alerts are ignored	**Cloud Monitoring alerting policies + SLOs**
Trend reports and predictions	**BigQuery + Agent Platform**
Protect PHI	**BAA, CMEK, VPC-SC, Sensitive Data Protection**

Exam trap: Rebuilding insurer integrations on day one. The case says legacy interfaces must keep working during migration.

Use gen AI to fix the catalog and the shopping experience

What they need

- Large product catalog maintained by hand
- Inconsistent attributes, descriptions and images
- Customers struggle to find the right product
- High call-center volume and cost
- Modernize on-prem databases and VMs
- Responsible, brand-safe AI output

Requirement	Google Cloud answer
Generate descriptions and attributes	**Gemini on Agent Platform, batch jobs**
Create or improve product images	**Imagen on Agent Platform**
Natural-language product search	**Vertex AI Search for commerce / RAG**
Self-service support	**Conversational agent + human handoff**
Relational catalog data	**Cloud SQL or AlloyDB**
Sales and catalog analytics	**BigQuery**
Quality and brand safety	**Human review, evaluation, grounding**

Exam trap: Auto-publishing generated content. Expect a review step and grounding in real catalog data.

Add gen AI to a cloud-native media platform

What they need

- Large library: podcasts, interviews, news, documentaries
- Already on GKE, Cloud Storage, BigQuery, Cloud Run functions
- Summaries and metadata for content at scale
- Content moderation and personalized recommendations
- Natural-language interaction and self-service support
- Lower storage cost, better ops and CI/CD

Requirement	Google Cloud answer
Summarize and tag audio / video	**Gemini multimodal (+ Speech-to-Text)**
Recommendations	**Embeddings + vector search, BigQuery**
Moderation	**Gemini safety filters, Model Armor, human review**
Viewer and support assistant	**Agent built with ADK on Agent Engine**
Cold media cost	**Cloud Storage Autoclass**
Per-team pipelines	**Central Cloud Build + Cloud Deploy**
Weak alerting	**Cloud Monitoring alerting + SLOs**

Exam trap: Re-platforming what already works. The right answer extends the existing GKE / BigQuery estate.

Connected, autonomous vehicles on a hybrid platform

What they need

- Self-driving and electric vehicle programs
- Heavy AI/ML: training, simulation, testing
- Legacy on-prem systems and plants
- Intermittent vehicle connectivity
- Inconsistent dealer and customer experience
- Data sovereignty, security, new data revenue

Requirement	Google Cloud answer
Vehicle telemetry ingestion	**Partner MQTT broker → Pub/Sub → Dataflow**
Telemetry storage and analytics	**Bigtable + BigQuery**
AV training and simulation	**GPUs / TPUs on GKE or Agent Platform**
Plant / edge and sovereign workloads	**Google Distributed Cloud**
In-car and dealer assistants	**Gemini agents on Agent Platform**
Partner and dealer APIs	**Apigee**
Data monetization	**BigQuery sharing (Analytics Hub)**
EU sovereignty and security	**Assured Workloads, Cloud EKM, SCC**

Exam trap: Choosing IoT Core. It is retired; use a partner MQTT broker feeding Pub/Sub.

How Cloud KMS keys are organized

Project
Isolate keys in a dedicated security project

Key ring
Location fixed at creation (region, multi-region, global)

Key
Purpose and protection level; rotation schedule

Key version
The actual material; one is primary

Encryption options: more control → more responsibility

Google default encryption
On by default, Google-managed keys, zero effort

CMEK · software keys
You own rotation, IAM, disable/destroy in Cloud KMS

CMEK · Cloud HSM
Keys in FIPS 140-2 Level 3 validated HSMs

CMEK · Cloud EKM
Keys stay in an external key manager outside Google

Envelope encryption: data is encrypted with a data encryption key (DEK); your Cloud KMS key (KEK) wraps the DEK.

Making CMEK work, and what breaks it

1

Grant the service agent

Give the service's agent (e.g. BigQuery, Cloud Storage, Cloud SQL) roles/cloudkms.cryptoKeyEncrypterDecrypter on the key, not your users.

2

Match locations

The key ring location must match or cover the resource location; a us-central1 key cannot protect a europe-west1 bucket.

3

Rotate

Rotation creates a new primary version for new writes. Older data keeps using older versions; it is not re-encrypted automatically.

4

Revoke = crypto-shred

Disabling or destroying a key version makes data unreadable. Destruction is scheduled (30 days by default) as a safety window.

5

Enforce at scale

Org policies restrict non-CMEK services and which projects keys can come from; Cloud KMS Autokey creates keys on demand.

6

Separate duties

Keep keys in a dedicated project; key admins cannot read data, data users cannot manage keys.

Secrets are not keys

Secret Manager essentials

- Stores API keys, passwords, certificates as versioned secrets
- Access with roles/secretmanager.secretAccessor per secret
- Pin a version or use “latest” in Cloud Run / GKE
- Rotation schedules send Pub/Sub notifications to a rotator
- Replication: automatic, or user-managed for data residency
- Secrets can be encrypted with CMEK; access is audit-logged

	Cloud KMS	Secret Manager
Stores	Cryptographic keys	Secret values (strings, blobs)
Can you read the value?	No: keys never leave KMS	Yes, with accessor role
Typical use	Encrypt data, CMEK, signing	DB passwords, API tokens
Versioning	Key versions, primary	Secret versions, enable / disable
Exam reflex	“Control encryption keys”	“Stop hard-coding credentials”

Pick the answer in one read

Scenario	Best answer
Regulator requires you to control and revoke keys for BigQuery and Cloud Storage	**CMEK with Cloud KMS, service agent granted Encrypter/Decrypter**
Keys must be in FIPS 140-2 Level 3 hardware	**Cloud HSM protection level**
Keys must never be held by Google	**Cloud EKM with an external key manager**
Every new resource in the org must use CMEK	**Org policy restricting non-CMEK services (+ Autokey)**
Cloud Run app has a database password in an env file	**Secret Manager, mounted or referenced by version**
Secrets must stay in the EU	**User-managed replication with EU locations only**
Credentials must rotate every 30 days	**Secret rotation schedule → Pub/Sub → rotation function**

EXAM DAY

Requirements first, managed second, cost always

Re-read both case studies on the split screen · Underline the hard constraint · Eliminate answers that add operations · For AI, ground before you tune · For security, CMEK for key control, Secret Manager for credentials